METHODOLOGY REPORT

Sepsis Onset Prediction Under Label Noise

PhysioNet Sepsis Challenge 2019 (Training Set A)

Task: ITSOLERA Task 1
Date: June 2026
Dataset: PhysioNet / Sepsis Challenge 2019 — Training Set A
Model: Calibrated XGBoost (Sigmoid Calibration, 3-Fold CV)

1. Executive Summary

This report describes a complete machine learning pipeline for predicting sepsis onset 6 hours before a physician's recorded diagnosis in ICU patients. The solution uses the PhysioNet Sepsis Challenge 2019 Training Set A, comprising 7,569 patient stays with 42 clinical features recorded at hourly intervals.

The core challenges addressed are: (1) high missingness (up to 60% per feature), (2) irregular time-series with non-uniform observation patterns, (3) noisy and delayed physician labels, and (4) strict prevention of temporal data leakage. The final model is a Sigmoid-calibrated XGBoost classifier achieving a Test ROC-AUC of 0.800 and Test AUPRC of 0.103 on a held-out patient cohort.

2. Dataset Overview

2.1 Source

Dataset: PhysioNet Sepsis Challenge 2019 — Training Set A

URL: https://physionet.org/content/challenge-2019/1.0.0/

Format: Per-patient pipe-separated value (.psv) files, one row per hour of ICU stay.

2.2 Key Statistics

Statistic	Value
Total patient files	7,569
Total hourly rows	293,739
Raw features	42 (vitals + labs + demographics + SepsisLabel)
Overall sepsis rate (row-level)	~2.1%
Features dropped (>95% missing in train)	11
Features retained after dropping	31

2.3 Features

The 42 raw features span four domains:

- Vital signs (7): HR, O2Sat, Temp, SBP, MAP, DBP, Resp
- Lab results (26): HCO3, FiO2, pH, PaCO2, SaO2, AST, BUN, Alkalinephos, Creatinine, Bilirubin (direct/total), Glucose, Lactate, Magnesium, Phosphate, Potassium, Bilirubin_total, TroponinI, Hct, Hgb, PTT, WBC, Fibrinogen, Platelets, etc.
- Demographics (5): Age, Gender, Unit1, Unit2, HospAdmTime
- Time index (1): ICULOS (ICU length-of-stay in hours)

- Label (1): SepsisLabel (0/1 — physician-recorded, noisy)

3. Methodology

3.1 Patient-Level Data Split (Leakage Prevention)

All splitting was performed at the patient level, not the row level, to prevent temporal data leakage across train/validation/test boundaries. Each ICU stay belongs exclusively to one split.

Split	Patients	Rows
Train	5,298 (70%)	204,828
Validation	1,135 (15%)	44,095
Test	1,136 (15%)	44,816

Stratification was applied on patient-level sepsis label (ever-septic vs. never-septic) to maintain class balance across splits. Set disjointness was verified programmatically.

3.2 Handling High-Missingness Time-Series

3.2.1 Column Dropping

Features with more than 95% missing values in the training set were dropped entirely. This threshold was computed on training data only to prevent information leakage. 11 features were removed:

- EtCO2, AST, Alkalinephos, Calcium, Bilirubin_direct, Lactate, Phosphate, Bilirubin_total, TroponinI, PTT, Fibrinogen

31 features were retained after this step.

3.2.2 Missingness Indicator Features

For all 24 clinical (non-demographic) features, a binary indicator feature was added (feature_name_missing = 1 if the original observation was absent). These indicators allow the model to distinguish truly-normal values from imputed values, and capture the clinical signal that a lab was not ordered.

3.2.3 Per-Patient Forward Fill

Within each patient stay, missing values were forward-filled in chronological order (ICULOS ascending). This propagates the last known measurement forward in time, mimicking clinical practice where the most recent observation is used until a new one is recorded.

3.2.4 Median Imputation for Remaining NaNs

Any value still missing after forward fill (because no prior observation existed for that patient and feature) was filled with the training-set median. Medians were computed exclusively on the training split and then applied to validation and test sets — no leakage.

After all imputation steps, zero missing values remained across all three splits.

3.3 Temporal Feature Engineering

3.3.1 Time Since Last Observation

For the 7 key vitals (HR, O2Sat, Temp, SBP, MAP, DBP, Resp), a time_since_last (TSL) feature was engineered: the number of hours elapsed since the last non-missing measurement within the same patient stay. This captures measurement staleness, which is a clinically meaningful signal (e.g., a temperature not taken for 6 hours is more uncertain than one from 1 hour ago).

3.3.2 Rolling Window Statistics

6-hour rolling mean and rolling standard deviation were computed for the 7 key vitals within each patient stay (minimum 1 period). These features capture trend and variability, both of which are important early signals of physiological deterioration leading to sepsis.

After feature engineering, the final feature matrix contained 73 features per row.

3.4 Label Noise Detection and Mitigation

Physician sepsis diagnoses in MIMIC-based datasets are known to suffer from two types of noise:

- Late recording: the diagnosis is documented hours after clinical onset
- Inconsistent recording: physicians vary in their adherence to Sepsis-3 criteria

3.4.1 Label Transition Flagging

A label_shift column was computed as the first-difference of SepsisLabel within each patient stay. Rows where |label_shift| = 1 (i.e., the exact transition from non-septic to septic) represent the most temporally uncertain labels, as the true onset is somewhere in the boundary window.

3.4.2 Sample Weighting for Noise Mitigation

Transition rows were assigned a sample weight of 0.5 (versus 1.0 for all other rows). During model training, this down-weights the influence of the most label-uncertain observations without entirely discarding them. This is a pragmatic, computationally lightweight noise-mitigation strategy that avoids the complexity of full label-noise models while acknowledging the structural uncertainty.

3.5 Model Architecture and Training

3.5.1 Baseline — Random Forest

A Random Forest (200 trees, max_depth=12, class_weight='balanced') was trained as a baseline with the transition-weighted samples. Validation ROC-AUC: 0.747, AUPRC: 0.064.

3.5.2 Candidate Models Compared

Model	Val ROC-AUC	Val AUPRC
Random Forest (200 trees, depth=12)	0.747	0.064
XGBoost (300 trees, depth=6, lr=0.05)	0.690	0.065

Model	Val ROC-AUC	Val AUPRC
XGBoost Tuned (500 trees, depth=4, lr=0.03)	0.744	0.076
Random Forest Deep (300 trees, depth=20)	0.760	0.064
Calibrated XGBoost (Sigmoid, 3-fold CV) ✓	0.754	0.078

3.5.3 Final Model — Calibrated XGBoost

The final model is a Sigmoid-calibrated XGBoost classifier (CalibratedClassifierCV, method='sigmoid', cv=3) wrapping:

- n_estimators=500, max_depth=4, learning_rate=0.03
- subsample=0.8, colsample_bytree=0.8, min_child_weight=5
- scale_pos_weight=46.28 (ratio of negatives to positives, handling class imbalance)
- eval_metric='aucpr' (area under precision-recall curve, appropriate for imbalanced labels)

Calibration was fitted inside 3-fold cross-validation on the training set to produce reliable probability estimates, which is important for downstream threshold selection and clinical decision-making.

3.6 Threshold Selection

Because sepsis prediction is a highly imbalanced classification problem (~2.1% positive rate), the default 0.5 probability threshold is inappropriate. Threshold selection was performed exclusively on the validation set by maximizing the F1 score across the precision-recall curve.

Optimal threshold (validation set): 0.0700

Validation performance at this threshold:

- F1 = 0.152
- Precision = 0.107
- Recall = 0.268

This threshold was then applied unchanged to the held-out test set, ensuring no information from the test set influenced the threshold choice.

4. Evaluation Results

4.1 Test Set Performance

Metric	Value
Test ROC-AUC	0.800
Test AUPRC	0.103
Threshold applied	0.070 (tuned on validation)

Metric	Value
Test Accuracy	0.94
Test F1 (sepsis class)	0.18
Test Precision (sepsis class)	0.13
Test Recall (sepsis class)	0.30
Test support (non-sepsis rows)	43,891
Test support (sepsis rows)	925

4.2 Confusion Matrix (Test Set)

At threshold 0.070:

	Predicted Non-Sepsis (0)	Predicted Sepsis (1)
Actual Non-Sepsis (0)	42,089 (TN)	1,802 (FP)
Actual Sepsis (1)	645 (FN)	280 (TP)

The confusion matrix reflects the inherent trade-off in imbalanced clinical prediction: at the chosen threshold, the model captures 30% of true sepsis cases (recall = 0.30) while maintaining an overall accuracy of 94%. Precision (0.13) reflects that approximately 1 in 8 positive predictions is a true sepsis case, consistent with a prevalence-adjusted early warning system.

4.3 Calibration Curve

A calibration curve (reliability diagram) was plotted using 10 probability bins on the test set. The model outputs were compared against observed sepsis frequencies. Sigmoid calibration via 3-fold cross-validation produced probability estimates substantially better aligned with empirical frequencies than the raw XGBoost outputs. The calibration curve shows the model is reasonably well-calibrated in the low-probability region (0.0–0.2), where the majority of predictions fall given the ~2% base rate.

Note: Plot files (confusion_matrix.png, calibration_curve.png) are saved as outputs in the accompanying notebook.

5. Temporal Leakage Prevention

Strict leakage prevention was enforced at every pipeline stage:

- Patient-level splits: no patient's rows appear in more than one split
- Column dropping threshold computed on training data only
- Missingness indicators derived from each split independently
- Forward fill performed within patient stays (no cross-patient leakage)

- Median imputation statistics fit on training set, applied to val/test
- Rolling statistics computed within patient stays (no future data used)
- Model calibration performed via cross-validation within training set
- Threshold selection performed exclusively on validation set
- Test set evaluated exactly once, at the end, with a frozen pipeline

6. Limitations and Future Work

6.1 Current Limitations

- Label noise mitigation is simple (transition down-weighting) and does not model the full distribution of labeling uncertainty
- Only Training Set A (7,569 patients) was used; Training Set B would nearly double the data and likely improve recall
- The 6-hour prediction window is approximated by the challenge labels rather than precisely shifted; a true shift would require applying a -6h label offset
- No sequence model (LSTM, GRU, Transformer) was used; the temporal structure is captured only via rolling features and TSL indicators
- AUPRC (0.103) is still low relative to the 2.1% base rate (random classifier AUPRC ≈ 0.021), indicating substantial room for improvement

6.2 Future Improvements

- Label smoothing or confident learning (CleanLab) for more principled noise handling
- LSTM or Temporal Convolutional Network to model the full time series
- Adding SOFA score components and antibiotic administration as features
- Ensembling the Random Forest and XGBoost outputs
- Using both PhysioNet Training Set A and B
- Hyperparameter optimization via Optuna

7. Reproducibility

7.1 Deliverables

- Notebook: ITSOLERA_Task__1_.ipynb (all steps, outputs, and plots)
- Report: This document (methodology_report.docx)
- Figures: confusion_matrix.png, calibration_curve.png (saved in notebook)

7.2 Dependencies

Library	Version Used	Purpose
Python	3.x	Runtime

Library	Version Used	Purpose
pandas	latest	Data loading, manipulation
numpy	latest	Numerical operations
scikit-learn	latest	RF, calibration, metrics
xgboost	latest	Gradient boosting model
matplotlib	latest	Plots
wget	system	Dataset download

7.3 Random Seed

All stochastic operations use random_state=42 for reproducibility.

7.4 Dataset Access

The PhysioNet Sepsis Challenge 2019 dataset is publicly available (no credentials required for Training Set A). Data is downloaded directly in the notebook using wget.

URL: https://physionet.org/content/challenge-2019/1.0.0/

8. References

[1] Reyna MA et al. Early Prediction of Sepsis from Clinical Data: The PhysioNet/Computing in Cardiology Challenge 2019. Critical Care Medicine, 2020.

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[5] Platt J. Probabilistic Outputs for Support Vector Machines. Advances in Large Margin Classifiers, 1999.